> quanta' which are localized at points in space, which move without dividing, and which can only be produced and absorbed as complete units.

These simple and clear lines are the real birth certificate of quantum theory. Note the wonderful initial 'It seems to me …', which recalls the 'I think …' with which Darwin introduces in his notebooks the great idea that species evolve, or the 'hesitation' spoken of by Faraday when introducing for the first time the revolutionary idea of magnetic fields. Genius hesitates.

The work of Einstein was initially treated by colleagues as the nonsensical juvenilia of an exceptionally brilliant youth. Subsequently it was for the same work that he received the Nobel Prize. If Planck is the father of the theory, Einstein is the parent who nurtured it.

But like all offspring, the theory then went its own way, unrecognized by Einstein himself. In the second and third decades of the twentieth century it was the Dane Niels Bohr who pioneered its development. It was Bohr who understood that the energy of electrons in atoms can only take on certain values, like the energy of light, and crucially that electrons can only 'jump' between one atomic orbit and another with fixed energies, emitting or absorbing a photon when they jump. These are the famous 'quantum leaps'. And it was in his institute in Copenhagen that the most brilliant young minds of the century gathered together to investigate and try to bring order to these baffling aspects of behaviour in the atomic world, and to build from it a coherent theory. In 1925 the equations of the theory finally appeared, replacing the entire mechanics of Newton.

It's difficult to imagine a greater achievement. At one stroke, everything makes sense, and you can calculate everything. Take one example: do you remember the periodic table of elements, devised by Mendeleev, which lists all the possible elementary substances of which the universe is made, from hydrogen to uranium, and which was hung on so many classroom walls? Why are precisely these elements listed there, and why does the periodic table have this particular structure, with these periods, and with the elements having these specific properties? The answer is that each element corresponds to one solution of the main equation of quantum mechanics. The whole of chemistry emerges from a single equation.

The first to write the equations of the new theory, basing them on dizzying ideas, would be a young German of genius, Werner Heisenberg.

Heisenberg imagined that electrons do not *always* exist. They only exist when someone or something watches them, or better, when they are